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Feature Selection Optimisation through Genetic Programming in Machine Learning Models

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Defini as perguntas de revisão, validei os termos e critérios de pesquisa, executei as pesquisas e verifiquei a triagem, revendo o título e o resumo de todos os estudos incluídos, dos registos excluídos na fase de elegibilidade e de uma amostra dos registos excluídos na fase de título.

Revi o texto final e assumo total responsabilidade por este documento.

Pedro Miguel Silva Roleira Marinho

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Chapter 1

Introduction

1.1 Context

Datasets used to train machine learning (ML) models frequently contain dozens, hundreds, or even thousands of features, although only part of them actually carries information about the target variable. Some application areas take this to the extreme. In the quantitative structure–activity relationship (QSAR) benchmarks studied by Wang et al. [166], which relate chemical compounds to their activity on drug targets, each sample is described by 1024 variables, while fewer than 700 samples are available. With so many candidate inputs, a model can easily latch onto misleading patterns, training becomes more expensive, and explaining how the model reaches its predictions becomes harder [27][45].

Feature selection addresses this issue by discarding the inputs that do not help and keeping those that do. When it is performed well, predictors can become more accurate, cheaper to train and to run, and easier to understand [45]. The literature usually distinguishes three ways of performing it [45][176]. *Filters* rate variables through statistical criteria computed from the data alone, before any model is involved, whereas *wrappers* repeatedly train a chosen model on different candidate subsets and keep the one under which it performs best. *Embedded* methods, in turn, choose variables while the model itself is being trained. In many practical pipelines, the selection relies on per-feature importance scores, which are nowadays often obtained from explainability tools such as SHapley Additive exPlanations (SHAP) [83][92][163] and Local Interpretable Model-agnostic Explanations (LIME) [122].

Evolutionary computation (EC) approaches the selection problem as a search. In a Genetic Algorithm (GA), each candidate solution can be represented as a string of bits stating which variables are kept, which makes encoding a subset straightforward [176][189]. Genetic Programming (GP) goes one step further, since it evolves the predictive model itself in the form of an expression tree, and therefore only the variables that end up inside the evolved expressions are effectively used [176][189]. As a result, selection and model construction happen within a single process, and the resulting expressions can be read and inspected [105]. Building on these ideas, recent studies have paired a GA, responsible for choosing the subset, with GP, responsible for learning from

it [127], while others have used Shapley values to decide which variables a GP system should retain [166]. Within this context, the present dissertation, carried out with Altice Labs, investigates how such evolutionary strategies can deliver feature selection for ML models that is both more effective and less costly.

1.2 Motivation

Ranking variables one by one assumes that the usefulness of each variable can be judged in isolation, an assumption that often fails in practice. Guyon and Elisseeff, for example, describe cases in which a variable with no predictive power on its own becomes informative once paired with another, as well as cases in which strongly correlated variables still add information to each other [45]. Similarly, permutation importance, which perturbs a single variable at a time, is known to misjudge variables that are correlated with others [166]. Consequently, when such rankings drive the selection, variables that only matter jointly risk being discarded, and the final model may lose both accuracy and credibility.

GP is appealing in this setting precisely because the expressions it evolves can combine variables explicitly. However, its implicit selection loses effectiveness as the number of candidate variables grows [27][166], and repeated runs may settle on different subsets [106]. Shapley values offer a principled way of measuring the contribution of each variable, because they average its effect over the possible combinations of the remaining variables [83]. Indeed, using them to filter the inputs of GP improved its generalisation and reduced the number of selected variables by at least 27% relative to standard GP [166]. Nevertheless, the main obstacle remains the computational cost. Since Shapley values require evaluating each variable across many combinations of the others, the selection stage itself becomes expensive, and on one of the QSAR datasets mentioned above the Shapley-based pipeline took roughly five times longer than standard GP [166]. Nesting complete GP runs inside a GA is also demanding, given that every candidate subset triggers a full evolutionary search [127]. Reducing this cost without losing the ability to capture interacting variables would therefore benefit anyone who needs small, accurate, and understandable models, such as data scientists working with wide datasets in industry.

1.3 Problem

Feature selection currently involves a compromise between efficiency and the ability to account for interactions between variables. On the one hand, ranking-based methods are fast, yet they can discard variables whose contribution only becomes visible in combination with others [45][166]. On the other hand, the evolutionary alternatives that do account for such interactions are expensive, either because they repeat a full GP search for every candidate subset [127] or because they compute Shapley values in a dedicated and time-consuming phase before learning [166]. Moreover, work combining Shapley values with GP has concentrated on symbolic regression, a setting in which the evolved expression is itself the predictor [102][166], and the same holds for recent attempts

to feed importance information into the evolutionary operators [187]. It therefore remains unclear how well these ideas carry over to classification and whether the subsets they produce also benefit other kinds of ML models. In fact, Wang et al. themselves identify the integration of the Shapley computation into the evolution, the reduction of its cost, and its application to classification as directions for further work [166].

Resolving this compromise would be valuable in practice for teams that must reduce high-dimensional data under limited computing resources, since a selection procedure that keeps jointly relevant variables without an excessive runtime would lead to models that are smaller, more accurate, and easier to interpret.

1.4 Research Questions

Based on the gaps described above, this work is guided by the following research questions:

RQ1 How can Shapley-value information be integrated into the GP evolutionary process, and how does this affect the computational cost of feature selection compared with two-stage and nested evolutionary approaches?

RQ2 Does this integration preserve the quality of the selection, in terms of predictive performance, subset size, stability across runs, and the recovery of features that are only informative in combination, when compared with those approaches and with rankings based on individual feature importance?

RQ3 To what extent are the selected subsets useful beyond GP itself, that is, when used to train other families of ML models, such as linear models, tree ensembles, and neural networks?

1.5 Hypothesis

Integrating Shapley-based importance into the GP evolutionary process reduces the computational cost of feature selection relative to two-stage and nested evolutionary approaches, while producing feature subsets that are at least as compact and accurate, and that remain useful when other families of ML models are trained on them.

Chapter 2

Literature Review

The research questions in Chapter 1 depend on what is already known about feature selection with GP, about Shapley values as a selection criterion, and about how both can be combined. This chapter explains how the studies that answer these questions were found and selected, in enough detail for another researcher to repeat the search and obtain a similar set. Section 2.1 justifies the type of review and Section 2.2 states the questions it answers. Sections 2.3 to 2.5 describe the search, the eligibility criteria and the selection procedure, Section 2.6 reports the outcome, and Section 2.7 discusses the limits of the process. The content of the selected studies is analysed in the State of the Art chapter.

2.1 Review Type

Literature reviews range from narrative, exploratory reviews to fully systematic reviews with formal quality appraisal and meta-analysis [44]. The choice of type is a trade-off between speed, replicability, comprehensiveness and bias [155]. This work adopts a *semi-systematic* review. The identification of studies is systematic: the databases, search strings, time window and eligibility criteria were fixed before the searches were run, and the flow of records is reported following the PRISMA 2020 statement [116]. Because part of the literature on Shapley values appears in machine-learning venues that the bibliographic databases index only partially, the database search is complemented with backward snowballing [172] and with the references recommended in the dissertation proposal and by the supervisor.

The review is not a full systematic review in the sense of Kitchenham and Charters [70]. It has a single reviewer and no formal quality scoring, and it does not include a meta-analysis, because the studies in scope report heterogeneous outcomes (different datasets, learners and metrics) that cannot be pooled. The supervisor was informed of this choice.

2.2 Review Questions

The review is driven by literature review questions (LRQs) derived from the research questions of this dissertation (Section 1.4). Each LRQ identifies a body of knowledge that one or more research questions depend on (Table 2.1).

LRQ1 Which genetic programming (GP) approaches to feature selection exist, and how do they estimate the relevance of features from the evolved population?

LRQ2 How have Shapley values, and game-theoretic attributions in general, been used as a feature-selection criterion, and which limitations have been reported?

LRQ3 Which approaches use feature-importance information *inside* an evolutionary search (e.g. to guide initialisation, terminal sampling, crossover or mutation), instead of as a separate pre- or post-processing stage?

LRQ4 Which techniques reduce the cost of estimating Shapley values (model-specific, sampling-based or surrogate-based), and which of them could be applied to GP trees?

LRQ5 How are feature-selection methods evaluated, namely with benchmarks whose relevant and interacting features are known, and by transferring the selected subset to other learners?

Table 2.1: Mapping between literature review questions and research questions.

	RQ1	**RQ2**	**RQ3**
LRQ1	✓	✓	
LRQ2		✓	
LRQ3	✓		
LRQ4	✓		
LRQ5		✓	✓

2.3 Search Strategy

2.3.1 PICOC Decomposition

To turn the review questions into search strings, they were decomposed with the PICOC framework (Population, Intervention, Comparison, Outcome, Context) [119][70], shown in Table 2.2. Comparison and Outcome are used to judge relevance during screening but are not part of the search strings, as restricting the search by outcome would exclude studies that report the same outcome with different vocabulary.

Table 2.2: PICOC decomposition of the review questions.

Facet	Value
Population	Supervised learning problems (classification and regression) on tabular data, including high-dimensional data.
Intervention	Feature selection based on genetic programming; feature selection or feature relevance based on Shapley values; use of feature-importance information within evolutionary search.
Comparison	Other feature-selection approaches (filter, wrapper, embedded), permutation importance, and two-stage GP and Shapley pipelines. No comparator is required for inclusion.
Outcome	Predictive performance, size of the selected subset, recovery of truly relevant and interacting features, computational cost.
Context	Machine learning and evolutionary computation research.

2.3.2 Data Sources

Three bibliographic databases were searched: **Scopus**, as a broad multidisciplinary index that also covers the Springer venues where much of the GP literature appears (e.g. *Genetic Programming and Evolvable Machines*, EuroGP); **IEEE Xplore**, which covers the *IEEE Transactions on Evolutionary Computation* and the IEEE Congress on Evolutionary Computation; and the **ACM Digital Library** (ACM Full-Text Collection), which covers the Genetic and Evolutionary Computation Conference (GECCO). The databases were accessed through institutional subscriptions.

The protocol initially also included the Web of Science Core Collection. It was dropped because no institutional access to it was available at the time of the search. The expected loss is limited, since Scopus has broader coverage than Web of Science in most subject areas [93] and the two main evolutionary-computation publishers (IEEE and ACM) were searched directly. In fact, after deduplication the ACM Digital Library contributed a single record not already retrieved from Scopus.

Google Scholar was not used, since its ranking cannot be reproduced and it does not export large result sets.

2.3.3 Search Terms and Strings

The search terms were organised in eight blocks (Table 2.3), with the terms of a block joined by `OR` and the asterisk used as the truncation wildcard.

Combining the blocks with `AND` gave the four search strings in Table 2.4, where "TAK" means that a block is matched against title, abstract and keywords, and "T" against the title only.

No search string targets LRQ4 directly. A database query on Shapley-value estimation returns thousands of explainability application papers, while the methodological core of this topic is concentrated in a small number of highly cited works; it is therefore covered through snowballing and recommendations (Section 2.5.4). LRQ5 is answered from the studies retrieved by the other strings, since evaluation design is described within them.

Table 2.3: Term blocks used to build the search strings.

Block	Concept	Terms (joined with OR)
B1	Genetic programming	`"genetic programming"`, `"gene expression programming"`, `"grammatical evolution"`
B2	Feature selection	`"feature selection"`, `"variable selection"`, `"attribute selection"`, `"feature subset"`
B3	Shapley values	`shapley`, `shap`
B4	Game-theoretic attribution	`"game theory"`, `"game theoretic"`, `"coalitional game*"`, `"cooperative game*"`
B5	Evolutionary computation	`"evolutionary computation"`, `"evolutionary algorithm*"`, `"genetic algorithm*"`, `"genetic programming"`, `"gene expression programming"`, `"grammatical evolution"`, `"particle swarm"`, `"differential evolution"`
B6	Feature selection or attribution	all terms of B2, `"feature importance"`, `"feature attribution"`, `"feature construction"`
B7	Feature importance signal	`"feature importance"`, `"variable importance"`, `"feature relevance"`, `"feature contribution*"`, `"feature frequenc*"`, `shapley`
B8	Evolutionary operators	`crossover`, `mutation`, `"genetic operator*"`, `"terminal set"`, `"terminal selection"`, `initialization`, `initialisation`

Table 2.4: Composition of the search strings and their purpose.

ID	Composition	LRQ	Purpose
QS1	B1 (TAK) AND B2 (TAK)	1, 5	Core body of work on feature selection with GP.
QS2	(B3 OR B4) (T) AND B2 (T)	2, 5	Shapley and game-theoretic feature selection. Restricted to titles because SHAP is used for post-hoc interpretation in a very large number of application papers, which mention both terms in the abstract without contributing to feature-selection methodology.
QS3	B5 (TAK) AND B3 (TAK) AND B6 (TAK)	2, 3	Hybrids of evolutionary computation and Shapley values. B5 is broader than B1 because such hybrids are scarce in GP alone. B4 is not used, to avoid retrieving evolutionary game theory.
QS4	B1 (TAK) AND B7 (TAK) AND B8 (TAK)	3	GP variants whose operators or initialisation are guided by feature importance, which is the mechanism this dissertation builds on.

As an example, QS1 in Scopus syntax is:

```
TITLE-ABS-KEY("genetic programming" OR "gene expression programming" OR
    "grammatical evolution")
AND TITLE-ABS-KEY("feature selection" OR "variable selection" OR
    "attribute selection" OR "feature subset")
AND PUBYEAR > 2014 AND PUBYEAR < 2027
AND (LIMIT-TO(DOCTYPE,"ar") OR LIMIT-TO(DOCTYPE,"cp")
     OR LIMIT-TO(DOCTYPE,"re") OR LIMIT-TO(DOCTYPE,"ch"))
AND LIMIT-TO(LANGUAGE,"English")
```

The exact strings for the three databases are listed in Appendix A. They were generated from the same blocks; only the field syntax differs (`TITLE-ABS-KEY`/`TITLE` in Scopus, `All Metadata`/`Document Title` in IEEE Xplore, `Abstract`/`Title` in the ACM Digital Library). In IEEE Xplore, the publication year (2015–2026) and content type (conferences, journals, books and early-access articles) were applied as result filters. In the ACM Digital Library, the strings were entered through the advanced-search form, one row per block, with the date range set in the form.

Refinement rule. To keep screening feasible while making the refinement itself replicable, the following rule was fixed before running the searches. *R1: if a string returns more than 500 records in Scopus, its most specific block (B2 in QS1, B3 in QS3, B7 in QS4) is restricted to the title, and the same restriction is applied to that string in every database.* No string reached the threshold (Table 2.5), so the strings were used as defined.

Table 2.5: Search refinement log (Scopus runs).

String	Date	Records	Action
QS1	2 Nov 2026	378	Below threshold; R1 not applied.
QS2	2 Nov 2026	208	Below threshold; R1 not applied.
QS3	2 Nov 2026	461	Below threshold; R1 not applied.
QS4	2 Nov 2026	5	Below threshold; R1 not applied.

2.3.4 Time Window

The database searches cover publications from January 2015 to the search date (2 November 2026). The survey by Xue et al. [176] reviews evolutionary-computation approaches to feature selection up to 2015 and acts as the entry point for earlier work; studies older than 2015 are admitted only through snowballing (criterion IC2 in Table 2.6).

2.4 Eligibility Criteria

Table 2.6 lists the inclusion (IC) and exclusion (EC) criteria and the stage at which each one is applied. A record is included only if it satisfies all inclusion criteria and none of the exclusion criteria. Every excluded record is annotated with the code of the *first* criterion that excludes it.

Table 2.6: Inclusion and exclusion criteria. Stage: S = search filter, D = deduplication, T = title screening, E = eligibility assessment on abstract and metadata, F = full-text reading (State of the Art phase).

Code	Criterion	Stage
IC1	Peer-reviewed journal article, conference paper, book chapter or review.	S
IC2	Published between 2015 and the search date. Not applied to records found by snowballing or recommendation.	S
IC3	Written in English.	S
IC4	Addresses feature selection or the estimation of feature relevance with GP, or with Shapley values or other game-theoretic attributions (including their use inside an evolutionary search), or the efficient estimation of Shapley values; or provides a survey or a benchmark resource for evaluating such methods. Feature selection with other evolutionary algorithms is in scope only when Shapley or game-theoretic information is part of the search.	T, E
IC5	Proposes, extends or empirically evaluates a method, or surveys or benchmarks such methods.	T, E
EC1	Duplicate of a record already in the pool, including a conference version superseded by a journal version that reports the same method and results.	D, E
EC2	Not a full paper: editorial, erratum, retracted paper, keynote, abstract, poster or extended abstract with fewer than four pages.	T, E
EC3	Full text not accessible through institutional subscriptions, open access, or a request to the authors.	F
EC4	Application study that uses GP, an evolutionary algorithm or SHAP only as an off-the-shelf tool, without a contribution to or evaluation of the feature-selection or attribution mechanism itself.	T, E
EC5	Shapley values used for a purpose other than feature relevance (e.g. data valuation, federated-learning incentives, network or energy cost allocation).	T, E
EC6	Feature selection is not the object of study: GP or the evolutionary algorithm is used only as the final learner or to tune hyper-parameters.	T, E
EC7	Non-tabular input (e.g. pixel or token attribution for images or text) with a method that does not transfer to tabular feature selection.	T, E

2.5 Selection Process

2.5.1 Record Management and Deduplication

All searches were run on 2 November 2026. The number of records of each string was recorded first (Table 2.7); the records were then exported with their abstracts (Scopus: RIS, all four strings combined with `OR`; IEEE Xplore: CSV per string; ACM Digital Library: BibTeX per string). Duplicates were removed with a script in two passes: first by exact match of the normalised DOI,

then by exact match of the normalised title (lower case, punctuation removed) and publication year; matches of the second pass were checked manually. Of the 273 duplicates, 20 were records retrieved by more than one string within Scopus (removed when the strings were combined), 250 were detected by DOI and 3 by title and year.

2.5.2 Title Screening

The 1146 unique records were first screened by title against IC4, IC5, EC2 and EC4–EC7. Records whose title did not allow a decision were kept for the next stage, so that uncertainty leads to over-inclusion rather than to the loss of relevant studies.

2.5.3 Eligibility Assessment

The records kept after title screening were assessed on their full abstract and bibliographic metadata. The abstract was used to decide IC4, IC5 and EC4–EC7; the document type and page range were used to apply EC2 (e.g. two-page conference posters); and records describing the same study were checked against EC1. Each decision and criterion code was recorded in a screening log, from which every number in the PRISMA diagram can be recomputed.

Full-text reading is part of the State of the Art phase. At that stage, EC3 (full-text access) will be applied and the decisions taken on the abstracts will be confirmed on the full texts; any study excluded then will be reported with its reason.

2.5.4 Snowballing and Other Sources

One iteration of backward snowballing [172] was performed on the two references of the dissertation proposal that are closest to the research questions, GPSHAP [166] and SLUG [127]. Their reference lists (28 and 55 distinct entries) contained 74 references that were not already in the pool; these were screened with the same criteria as the database records, except IC2.

In addition, the four references recommended for this work were considered: the three references of the dissertation proposal [166][127][106] and the TreeSHAP paper [84], recommended by the supervisor. Two of them had already been retrieved by the database search; the other two were added.

Records obtained in this way are reported separately (“other methods”) in the PRISMA diagram (Figure 2.1).

2.5.5 Deviations from the Protocol

Five changes were made to the protocol after it was defined. Three of them narrow the process and are described where they apply: Web of Science was not searched (Section 2.3), full-text eligibility was moved to the State of the Art phase, and snowballing was limited to one backward iteration on two start papers, without forward snowballing.

The other two concern IC4. First, the search returned a cluster of studies on feature (terminal) selection in GP hyper-heuristics for scheduling and routing. These problems are not supervised learning, but the studies estimate the relevance of features from the evolved population, for example from how often a feature appears in good individuals, which is the subject of LRQ1 and LRQ3; they were therefore admitted. Second, the wording of IC4 was made explicit about two cases: feature selection with evolutionary algorithms other than GP is in scope only when Shapley or game-theoretic information is part of the search, as the PICOC intervention requires, and benchmark resources are in scope because LRQ5 needs them.

2.5.6 Use of AI Tools

A generative AI assistant (Claude Opus 5.5, Anthropic) supported the preparation of this chapter. It helped draft the protocol and the text, generated the database-specific search strings from the term blocks, wrote the deduplication script and produced a first screening of the records. It was also used for language revision, to improve the clarity of the text and to translate passages between Portuguese and English. The author defined the review questions, validated the search terms and the eligibility criteria, executed the searches, and reviewed the screening by checking the title and abstract of every included study, of the records excluded at the eligibility stage and of a random sample of the records excluded at the title stage. The author revised the final text and is responsible for all decisions reported in this chapter.

2.6 Results of the Selection Process

Table 2.7 shows the number of records returned by each string in each database on 2 November 2026.

Table 2.7: Records retrieved per database and search string.

Database	**QS1**	**QS2**	**QS3**	**QS4**	**Total**
Scopus	378	208	461	5	1052
IEEE Xplore	157	61	111	8	337
ACM DL	18	6	5	1	30
Total	553	275	577	14	1419

Of the 1419 records retrieved, 273 were duplicates, leaving 1146 records for title screening. Of these, 938 were excluded (EC4: 800, IC4: 65, EC6: 44, EC7: 28, EC2: 1); the large share of EC4 reflects the many application studies that combine a metaheuristic-tuned model with SHAP-based interpretation, retrieved mostly by QS3. The remaining 208 records were assessed on their abstracts and metadata, and 38 were excluded (EC4: 17, IC4: 16, EC2: 2, EC1: 1, EC6: 1, EC7: 1), leaving 170 studies from the databases. Snowballing and recommendations added 12 studies, for a total of 182 included studies. Figure 2.1 shows the complete flow.

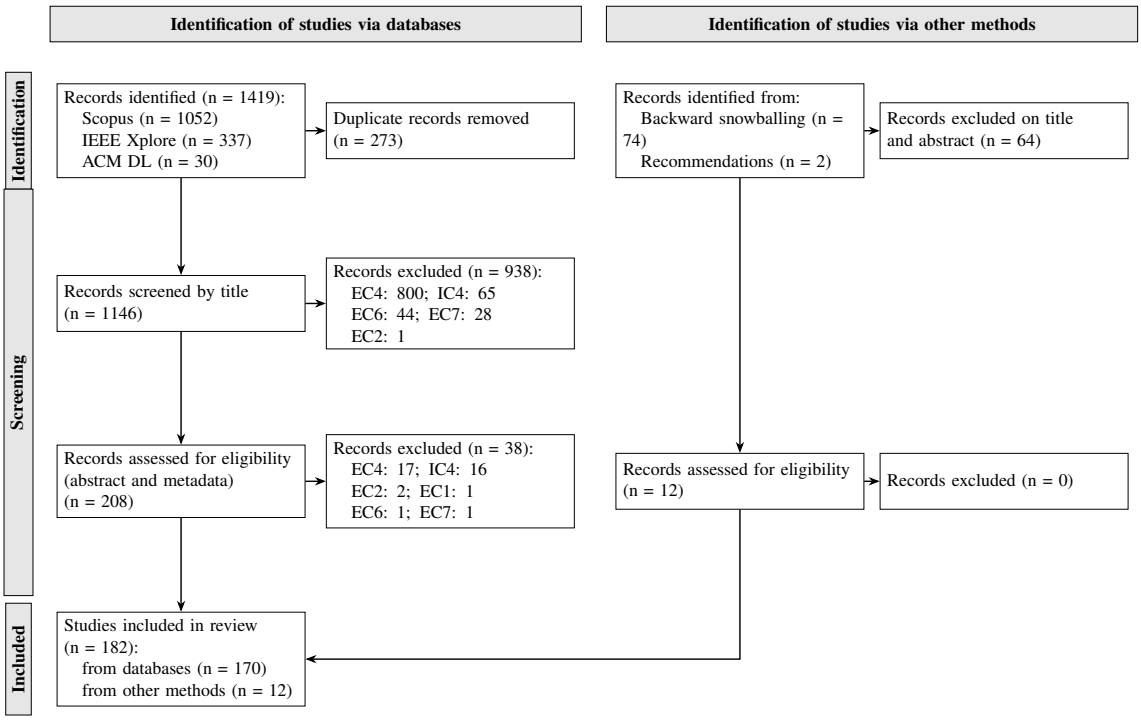


Figure 2.1: PRISMA 2020 flow diagram of the selection process, adapted from [116].

Table 2.8 groups the included studies by the review question they mainly inform, based only on their scope. Most of them concern GP-based feature selection (G1, 90 studies) or Shapley-based feature selection (G2, 60), while only 17 use importance or Shapley information *inside* an evolutionary search (G3), which is the mechanism RQ1 investigates. This imbalance is a first indication of the gap this dissertation addresses, to be confirmed by the analysis in the State of the Art chapter. All included studies are listed in the Bibliography.

Table 2.8: Included studies grouped by the review question they mainly inform, in chronological order within each group.

Group	Studies	n
G1 – GP-based feature selection, including GP hyper-heuristics (LRQ1)	[148][72][34][131][18][63][147][5][26][99][158][27][31][32] [82][100][113][117][149][159][178][17][91][164][8][53][11] [23][106][182][185][54][56][57][58][88][87][108][115][170] [49][94][138][179][183][186][188][10][59][80][126][130] [139][175][21][50][68][140][144][145][146][6][52][13][14] [35][78][89][127][156][190][25][37][51][60][73][76][110] [114][128][137][169][174][55][15][24][28][134][177][184]	90
G2 – Shapley and game-theoretic feature selection (LRQ2)	[121][42][103][104][171][150][2][157][30][92][161][33][39] [19][43][65][180][66][152][163][168][36][61][67][69][79] [90][98][101][102][129][136][167][165][16][38][47][48][111] [125][124][132][143][142][153][192][166][191][7][20][41] [71][81][85][86][97][118][123][133][160]	60

Group	Studies	n
G3 – Importance or Shapley information inside evolutionary search (LRQ3)	[22][12][107][135][181][64][141][77][4][40][173][3][95][187] [62][96][154]	17
G4 – Estimation of Shapley values and feature relevance (LRQ4)	[151][83][29][84][9][1]	6
G5 – Surveys, benchmarks and evaluation resources (LRQ5)	[46][162][176][112][74][75][109][189][120]	9
Total		**182**

2.7 Threats to Validity

Selection bias. The screening relied on titles and abstracts and on a first pass by an AI assistant, reviewed by a single reviewer, so individual decisions may still be wrong in either direction. Keeping every record whose title did not allow a decision, and logging the reason for every exclusion, makes such errors visible and correctable during the full-text reading.

Scope of the included set. IC4 admits Shapley-based feature selection methods that were proposed for a specific domain when they contribute a new selection mechanism, which explains the size of group G2; studies that only apply a standard SHAP ranking were excluded (EC4). The boundary between the two cases is a judgement and is the most likely source of disagreement between reviewers.

Database coverage. Without Web of Science, and with NeurIPS and ICML proceedings only partially indexed, some Shapley-value methods may have been missed. Snowballing reduces this risk only in part, as it covered two start papers and one backward iteration; forward snowballing is planned for the State of the Art phase.

Search terms. Studies that describe feature selection with different vocabulary (e.g. “input selection” or “dimensionality reduction”) may be missed, as may Shapley-based feature-selection studies whose title does not name both concepts (QS2).

Time window and language. Restricting the database search to 2015 onwards and to English may exclude relevant work; older foundational studies were admitted through snowballing.

2.8 Summary

A semi-systematic review was defined and applied: five review questions derived from the research questions, four search strings over three databases, explicit eligibility criteria, a two-stage

screening and backward snowballing. It produced 182 studies, grouped in Table 2.8. The State of the Art chapter will read these studies in full, confirm their eligibility and analyse them group by group, in particular to establish whether the gap suggested by the small size of G3 holds.

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Appendix A

Search Strings per Database

This appendix lists the exact search strings executed in each database on 2 November 2026. All strings are derived from the same term blocks (Table 2.3); only the field syntax changes between databases. IEEE Xplore and the ACM Digital Library do not support wildcards inside quoted phrases, so those phrases were expanded into their singular and plural forms. Rule R1 (Section 2.3) was not triggered by any string. In the ACM Digital Library the strings were entered through the advanced-search form (one row per block); the query below is the one displayed by the library on the results page.

A.1 Scopus (Advanced search)

QS1

```
TITLE-ABS-KEY("genetic programming" OR "gene expression programming" OR
    "grammatical evolution")
AND TITLE-ABS-KEY("feature selection" OR "variable selection" OR
    "attribute selection" OR "feature subset")
AND PUBYEAR > 2014 AND PUBYEAR < 2027
AND (LIMIT-TO(DOCTYPE,"ar") OR LIMIT-TO(DOCTYPE,"cp")
     OR LIMIT-TO(DOCTYPE,"re") OR LIMIT-TO(DOCTYPE,"ch"))
AND LIMIT-TO(LANGUAGE,"English")
```

QS2

```
TITLE(shapley OR shap OR "game theory" OR "game theoretic" OR
    "coalitional game*" OR "cooperative game*")
AND TITLE("feature selection" OR "variable selection" OR
    "attribute selection" OR "feature subset")
AND PUBYEAR > 2014 AND PUBYEAR < 2027
AND (LIMIT-TO(DOCTYPE,"ar") OR LIMIT-TO(DOCTYPE,"cp")
     OR LIMIT-TO(DOCTYPE,"re") OR LIMIT-TO(DOCTYPE,"ch"))
AND LIMIT-TO(LANGUAGE,"English")
```

QS3

```
TITLE-ABS-KEY("evolutionary computation" OR "evolutionary algorithm*" OR
    "genetic algorithm*" OR "genetic programming" OR
    "gene expression programming" OR "grammatical evolution" OR
    "particle swarm" OR "differential evolution")
AND TITLE-ABS-KEY(shapley OR shap)
AND TITLE-ABS-KEY("feature selection" OR "variable selection" OR
    "attribute selection" OR "feature subset" OR "feature importance" OR
    "feature attribution" OR "feature construction")
AND PUBYEAR > 2014 AND PUBYEAR < 2027
AND (LIMIT-TO(DOCTYPE,"ar") OR LIMIT-TO(DOCTYPE,"cp")
     OR LIMIT-TO(DOCTYPE,"re") OR LIMIT-TO(DOCTYPE,"ch"))
AND LIMIT-TO(LANGUAGE,"English")
```

QS4

```
TITLE-ABS-KEY("genetic programming" OR "gene expression programming" OR
    "grammatical evolution")
AND TITLE-ABS-KEY("feature importance" OR "variable importance" OR
    "feature relevance" OR "feature contribution*" OR "feature frequenc*" OR
    shapley)
AND TITLE-ABS-KEY(crossover OR mutation OR "genetic operator*" OR
    "terminal set" OR "terminal selection" OR initialization OR
    initialisation)
AND PUBYEAR > 2014 AND PUBYEAR < 2027
AND (LIMIT-TO(DOCTYPE,"ar") OR LIMIT-TO(DOCTYPE,"cp")
     OR LIMIT-TO(DOCTYPE,"re") OR LIMIT-TO(DOCTYPE,"ch"))
AND LIMIT-TO(LANGUAGE,"English")
```

A.2 IEEE Xplore (Command search)

QS1

```
("All Metadata":"genetic programming" OR
    "All Metadata":"gene expression programming" OR
    "All Metadata":"grammatical evolution")
AND ("All Metadata":"feature selection" OR
    "All Metadata":"variable selection" OR
    "All Metadata":"attribute selection" OR "All Metadata":"feature subset")
[Filters: Year 2015-2026; Conferences, Journals, Books, Early Access Articles]
```

QS2

```
("Document Title":shapley OR "Document Title":shap OR
    "Document Title":"game theory" OR "Document Title":"game theoretic" OR
```

```
    "Document Title":"coalitional game" OR
    "Document Title":"coalitional games" OR
    "Document Title":"cooperative game" OR
    "Document Title":"cooperative games")
AND ("Document Title":"feature selection" OR
    "Document Title":"variable selection" OR
    "Document Title":"attribute selection" OR
    "Document Title":"feature subset")
[Filters: Year 2015-2026; Conferences, Journals, Books, Early Access Articles]
```

QS3

```
("All Metadata":"evolutionary computation" OR
    "All Metadata":"evolutionary algorithm" OR
    "All Metadata":"evolutionary algorithms" OR
    "All Metadata":"genetic algorithm" OR
    "All Metadata":"genetic algorithms" OR
    "All Metadata":"genetic programming" OR
    "All Metadata":"gene expression programming" OR
    "All Metadata":"grammatical evolution" OR
    "All Metadata":"particle swarm" OR
    "All Metadata":"differential evolution")
AND ("All Metadata":shapley OR "All Metadata":shap)
AND ("All Metadata":"feature selection" OR
    "All Metadata":"variable selection" OR
    "All Metadata":"attribute selection" OR "All Metadata":"feature subset" OR
    "All Metadata":"feature importance" OR
    "All Metadata":"feature attribution" OR
    "All Metadata":"feature construction")
[Filters: Year 2015-2026; Conferences, Journals, Books, Early Access Articles]
```

QS4

```
("All Metadata":"genetic programming" OR
    "All Metadata":"gene expression programming" OR
    "All Metadata":"grammatical evolution")
AND ("All Metadata":"feature importance" OR
    "All Metadata":"variable importance" OR
    "All Metadata":"feature relevance" OR
    "All Metadata":"feature contribution" OR
    "All Metadata":"feature contributions" OR
    "All Metadata":"feature frequency" OR
    "All Metadata":"feature frequencies" OR "All Metadata":shapley)
AND ("All Metadata":crossover OR "All Metadata":mutation OR
    "All Metadata":"genetic operator" OR "All Metadata":"genetic operators" OR
    "All Metadata":"terminal set" OR "All Metadata":"terminal selection" OR
```

```
    "All Metadata":initialization OR "All Metadata":initialisation)
[Filters: Year 2015–2026; Conferences, Journals, Books, Early Access Articles]
```

A.3 ACM Digital Library (Advanced search, query as displayed)

QS1

```
[[Abstract: "genetic programming"] OR
    [Abstract: "gene expression programming"] OR
    [Abstract: "grammatical evolution"]]
AND [[Abstract: "feature selection"] OR [Abstract: "variable selection"] OR
    [Abstract: "attribute selection"] OR [Abstract: "feature subset"]]
AND [E-Publication Date: (01/01/2015 TO 12/31/2026)]
```

QS2

```
[[Title: shapley] OR [Title: shap] OR [Title: "game theory"] OR
    [Title: "game theoretic"] OR [Title: "coalitional game"] OR
    [Title: "coalitional games"] OR [Title: "cooperative game"] OR
    [Title: "cooperative games"]]
AND [[Title: "feature selection"] OR [Title: "variable selection"] OR
    [Title: "attribute selection"] OR [Title: "feature subset"]]
AND [E-Publication Date: (01/01/2015 TO 12/31/2026)]
```

QS3

```
[[Abstract: "evolutionary computation"] OR
    [Abstract: "evolutionary algorithm"] OR
    [Abstract: "evolutionary algorithms"] OR [Abstract: "genetic algorithm"] OR
    [Abstract: "genetic algorithms"] OR [Abstract: "genetic programming"] OR
    [Abstract: "gene expression programming"] OR
    [Abstract: "grammatical evolution"] OR [Abstract: "particle swarm"] OR
    [Abstract: "differential evolution"]]
AND [[Abstract: shapley] OR [Abstract: shap]]
AND [[Abstract: "feature selection"] OR [Abstract: "variable selection"] OR
    [Abstract: "attribute selection"] OR [Abstract: "feature subset"] OR
    [Abstract: "feature importance"] OR [Abstract: "feature attribution"] OR
    [Abstract: "feature construction"]]
AND [E-Publication Date: (01/01/2015 TO 12/31/2026)]
```

QS4

```
[[Abstract: "genetic programming"] OR
    [Abstract: "gene expression programming"] OR
    [Abstract: "grammatical evolution"]]
AND [[Abstract: "feature importance"] OR [Abstract: "variable importance"] OR
```

```
    [Abstract: "feature relevance"] OR [Abstract: "feature contribution"] OR
    [Abstract: "feature contributions"] OR [Abstract: "feature frequency"] OR
    [Abstract: "feature frequencies"] OR [Abstract: shapley]]
AND [[Abstract: crossover] OR [Abstract: mutation] OR
    [Abstract: "genetic operator"] OR [Abstract: "genetic operators"] OR
    [Abstract: "terminal set"] OR [Abstract: "terminal selection"] OR
    [Abstract: initialization] OR [Abstract: initialisation]]
AND [E-Publication Date: (01/01/2015 TO 12/31/2026)]
```